KUNAPULI VENKATA SAI KIRAN

Applied AI ML Lead, Vice President at JPMorgan Chase, Hyderabad, India

kirankunapuli@duck.com LinkedIn GitHub HuggingFace Kaggle **M** Medium

PROFILE

Experienced Data Scientist and AI/ML Leader with **13 years** in IT, **10+ years** in data science and **5 years** in technical leadership, driving foundation models, recommender systems, agentic GenAI and high-impact production ML.

PROFESSIONAL EXPERIENCE

JPMorgan Chase **Jul 2024 – Present**

Applied AI ML Lead (Vice President) *Hyderabad, India*

- Co-developed the **Chase Customer Foundational Model**, a transformer pre-trained on **52B+** card, deposit, digital and offer events for **95M+ customers** with masked, next-event, time-gap and contrastive losses on **24 H100 GPUs** (Ray, 3 nodes). Built training data and evaluated its 512-d embeddings on downstream tasks like L1 offer ranking.
- Re-architected **L1 offer ranking** with **8+** deep recommender and contextual-bandit models (DLRM, DCN-V2, SAINT, FT-Transformer, NeuralUCB, MoE), **doubling offline click-through rate** over LinUCB.
- Built a **multi-agent marketing assistant** (LangGraph, AWS Bedrock, agentic RAG) that turns business questions into SQL and model plans; cut discovery time **98%** and campaign time-to-market from **4 weeks to 5 days**.
- Built the **Account Origination Model** retraining pipeline on Databricks, MLflow and Unity Catalog, automating **monthly** releases across **49 products** with champion/challenger model promotion.
- Built Karpathy-inspired **AutoResearch** for LLM-driven experiment loops, adopted by other teams; contributed to **Casper**, a multi-agent AI assistant connecting Databricks, Jira, AWS and 6 more enterprise tools.

Tokopedia **Dec 2021 – Jul 2024**

Senior Lead Data Scientist *Hyderabad, India*

- Led the **Ads Ranking** and **Keyword Recommendation** projects for TopAds, Tokopedia's advertising platform, managing a team of **4 data scientists** from research to production.
- Developed the **Real-time Ads Ranking** system with a scalable inference pipeline serving **1K requests/s** at sub-80 ms latency, lifting CTR **3.25%** and ads revenue **1.57%** (1.6B IDR).
- Optimized inference of an existing deep learning model with post-training quantization, increasing revenue by **1.43%** (1.4B IDR monthly) and saving up to **166M IDR** in inference costs monthly.

ADP **Nov 2019 – Dec 2021**

Data Scientist *Hyderabad, India*

- Built multilingual **Job** and **PayCode Classification** models for ADP DataCloud using Language-agnostic BERT (LaBSE) and multilingual Universal Sentence Encoder (mUSE), cutting time-to-market from **6 to 2 months**.
- Deployed models on AWS (Step Functions, ECR, SageMaker, API Gateway) as batch transform pipelines and real-time endpoints, scaling model serving capacity **3x** to handle growing inference demand.

DBS Bank **Oct 2017 – Oct 2019**

Data Scientist *Hyderabad, India*

- Built a LightGBM **Cheque Risk Scoring** classifier for DBS Self-Service Banking that risk-scores every inward cheque, reducing false negatives by **18%** through effective class-imbalance handling.
- Built Prophet forecasts for ATMs and cash-handling devices to guide cash replenishment, cutting shortages by **15%**.

Tech Mahindra **Nov 2013 – Oct 2017**

Software Engineer *Hyderabad, India*

- Delivered **4 General Electric projects** from data ingestion (APIs, R, Talend) to ETL and predictive analytics.
- Engineered cash-collection models (clustering, decision trees) and ARIMA payroll forecasting for General Electric.

TECHNICAL SKILLS

ML & DL: RecSys, Ranking & Personalization, Contextual Bandits, Deep Learning, Embeddings, NLP, Forecasting

GenAI & Agents: LLMs, Foundation Models, Transformers, Fine-tuning (LoRA, PEFT), RAG, Multi-agent Systems, MCP

Languages & Frameworks: Python, R, SQL, PyTorch, TensorFlow, Hugging Face, scikit-learn, LangGraph, LangChain

Cloud, Data & MLOps: AWS (Bedrock, SageMaker), Databricks, Spark, Ray, DDP, MLflow, Airflow, Milvus, A/B Testing

EDUCATION

GITAM University, Hyderabad **2009 – 2013**

Bachelor of Technology (B.Tech) in Electronics and Communication Engineering *CGPA 8.4*

AWARDS & COURSES

2nd place, Continued Pre-training of LLM on Legal Contracts, Summer School on NLP & LLM, IIIT Hyderabad Jul 2024

Super Techie Award, DBS Bank: Pride Connect Award for Consumer Banking Analytics Dec 2018